Thomas Fieguth

Robotics & Mechatronics Engineer
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Skills

- **Programming:** C++, Python, Kotlin, Onshape FeatureScript · Linux, Git
- **Robotics:** ROS2, Nav2, sensor fusion (EKF), path planning & tracking, point clouds / ICP, Qt
- **Mechanical:** SolidWorks, Onshape, CNC (mill & lathe), machining, welding, soldering, woodworking
- **Languages:** English and German (fluent), French (immersion)

Experience

Werkr Tractors – Waterloo, ON — Jan 2024 – Present
Software Engineer (Full-time), May 2026 – Present · Software / Robotics Co-op, May – Aug 2025 and Jan – Apr 2024

- Designed the autonomous tractor's guidance system as C++ ROS2 lifecycle nodes: a mission-manager state machine, a speed-profiled trajectory planner, and a Stanley path tracker with curvature feedforward
- Tuned the Extended Kalman Filter fusing RTK GPS, GPS heading and onboard sensors; wrote Python rosbag tooling that found and fixed 3 estimator faults (GPS overconfidence, heading over-weighting, RTK yaw jumps)
- Added Nav2 obstacle handling: custom costmap layers, Smac Hybrid detours within the allowed row deviation, and the collision monitor as an independent safety stop
- Built the Qt/ROS2 operator interface and path generator for 5 mission types, using modified Dubins/Reeds–Shepp paths so the tractor never steers at a standstill
- Improved row-following performance through field testing with camera-based plant detection

GEI Consultants, Inc. – Kitchener, ON — May – Aug 2023
Project Designer Co-op

- Analyzed pump efficiency and friction losses at wastewater pumping stations, showing pumps ran at 93% of expected capacity and piping resistance was the main performance limit
- Built a parametric cost model for new water and wastewater treatment plants using only flow rate and processes
- Conducted site inspections and wrote formal client reports

Accelerated Systems Inc. – Waterloo, ON — Jan – Dec 2022
Mechanical / Software Engineer Co-op, Sep – Dec 2022 · Machinist & Technical Support Co-op, Jan – Apr 2022

- Built a Kotlin app to remotely drive an autonomous lawnmower and show live proximity, detection and GPS data
- Wrote C++ ROS2 nodes for proximity safety stops, geofencing and joystick teleoperation on the mower
- Designed a motor test stand and dynamometer for e-bike motors in SolidWorks
- Programmed CNC machines for prototype parts; machined and assembled golf cart motor casings

Projects

Depth-Sensing Woodcarving Guide – *Capstone, team of 4* — Sep 2025 – Apr 2026
Winner, N.S. Robertson Foundation Capstone Design Award

- Depth camera and projector rig that compares point clouds to the user's CAD model and projects a live depth-difference map onto the wood, showing where and how deep to carve next
- Did all CAD and mechanical manufacturing: 3 frame prototypes and 5 camera-projector mount designs
- Helped build and debug the camera-to-projector pipeline; implemented 3 iterations of the Python/Qt interface

Personal Projects

- **Farm Topography Relief Map:** wrote an Onshape FeatureScript tool that turns elevation data into nested, self-supporting CNC contour layers across 9 wood species
- **Catan River Tile Optimizer:** Monte Carlo search with DFS conflict resolution to find tile sets maximizing valid river layouts, plus procedural terrain generation (Python)
- **Stormbreaker Axe Replica:** load-bearing 1/4" steel skeleton under a sheet-metal shell; welded and heat-treated
- **CNC Wooden Globe:** 60 CNC-cut walnut and maple triangles assembled into a hollow, motor-driven globe

Education

BASc Systems Design Engineering – University of Waterloo — Graduated Apr 2026
Graduated with Dean's Honours · N.S. Robertson Foundation Capstone Design Award